Lecture 14 · Part III · Lecture notes

Detection and segmentation

Dataset: COCO, 128 images

How to use these notes

The primary text for this lecture is Géron, Chapter 12. These notes are not a replacement for it and do not repeat it: they are the lecture's own argument written out at length — the derivation done slowly, the numbers with the runs that produced them, and the reasoning between one slide and the next that a deck can only gesture at. Read the chapter for the method; read these for why the lecture makes the choices it does. Every figure quoted here is printed by this lecture's notebook, and the course checks that mechanically rather than trusting it.

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1 When a label is not an answer

Three lectures of vision, all answering the same question: what is in this image? Neither a convolutional network nor a pretrained backbone can express *where* the object is, or *how many* there are. A photograph with three cyclists has one label and three answers.

A retail shelf audit puts it plainly: *"We photograph shelves. We need to know how many of each product are on each shelf, and where each one is, so that a picking route can be planned. A list of the categories present is no use to us."*

	Classification	**Detection**
Question	what is in this picture	what is where, and how many
Output	one vector, length K	three arrays, length N
N is	fixed before you look	decided by the image and by two cut-offs
Wanted from pooling	invariance	equivariance

The last row is Lecture 12 with a consequence: *a network that has learned not to care where the object is cannot tell you where it is.* So the output stops being a label and becomes a set of boxes — which means the first thing we need is a new notion of what a correct answer is.

Failure condition — a box is four numbers, and which four is not agreed

`[x1, y1, x2, y2]` is two opposite corners, used by torchvision. `[x, y, w, h]` is a corner and a size, used by the COCO annotation file. `[cx, cy, w, h]` is a centre and a size, used by the YOLO family.
You will hold the first two in the same notebook. Read `[100, 50, 20, 30]` as corners and you get a box 20 wide when it should be 20 *starting at* 100 — a valid box, in the wrong place, with no error.

1.1 Our corpus is 128 images

Not COCO. 128 images out of the 5,000 in `val2017`, chosen as the numerically lowest image ids — a rule, not a selection. Full COCO with its training split is roughly 20 GB and is not downloaded anywhere in this course.

Quantity	Value
Images	128
Annotated objects, crowd regions excluded	898
Crowd regions, dropped	8
Distinct categories appearing	73 of 80
Objects per image — mean	7.02
Objects per image — median	5
Objects per image — range	1 to 29

Mean 7.02 against median 5: a long right tail, and one image alone carries 29 objects. And 39.0% of every annotated object in this corpus is a *person* — remember that when we start averaging over categories.

Every number in this lecture is measured on 128 images. When we quote a score, we quote the sample size beside it: a measurement quoted without its sample size is not a measurement.

2 Derivation: scoring a box, scoring a detector

2.1 One quantity, four requirements

Before writing anything down, fix what it has to do: be 1 for coincident boxes and 0 for disjoint ones; punish a box for being too large; punish a box for being too small; and be dimensionless, so a cup and a sofa are on one scale.

Requirement 2 says the predicted area must be in the denominator, requirement 3 says the true area must be too, and requirement 4 says the numerator must be an area as well. There is essentially one candidate:

$$\mathrm{IoU}(A, B) = \frac{|A \cap B|}{|A \cup B|} = \frac{|A \cap B|}{|A| + |B| - |A \cap B|}.$$

The second form is the one you compute — the union is never built — and subtracting the intersection is not an optimisation: add the two areas and you have counted the overlap twice. IoU is also called the **Jaccard index**; its range is $[0, 1]$, and it is exactly 1 only when $A = B$. The intersection of two axis-aligned boxes is another axis-aligned box, which is what makes it cheap: the overlap's top-left corner is the elementwise maximum of the two top-left corners, its bottom-right the elementwise minimum, and its width and height are clamped at zero.

2.2 IoU as a loss, and why it fails

Two boxes, each 100 by 100. Fix the first at the origin, slide the second right by d pixels, and ask autograd for the value and the gradient. Synthetic on purpose — nothing here depends on a dataset, so the conclusion is a property of the formula rather than of COCO. Past 100 pixels, where the boxes stop touching, IoU is zero and so is its gradient, at every separation.

Why the gradient is exactly zero, not merely small

For $d \geq 100$ the overlap width is $\max(0, 100 - d) = 0$, so

$$|A \cap B| \equiv 0 \implies \text{IoU} \equiv 0 \implies \frac{\partial\, \text{IoU}}{\partial d} \equiv 0.$$

The clamp makes the function *constant* on the whole disjoint region, not merely small there — and a constant has no descent direction. Not a weak one: none.

Failure condition — a loss that is flat where you need it

Take $\mathcal{L}_{\text{IoU}} = 1 - \text{IoU}(A, B)$ with B predicted — attractive, because it optimises the thing that is reported. But IoU is exactly zero for every disjoint pair, so a proposal that starts anywhere outside the object it should find receives the same loss and the same zero gradient wherever it is: nothing in the loss knows which way the object lies. The optimiser is not slow here — it has no direction to follow at all. This is a property of the formula, and no optimiser, learning rate or initialisation repairs it.

The repair is to put the empty space back in. With C the smallest box containing both,

$$\text{GIoU}(A, B) = \text{IoU}(A, B) - \frac{|C| - |A \cup B|}{|C|}.$$

When the boxes are disjoint and move apart, $|C|$ grows and $|A \cup B|$ does not, so the penalty grows — and the difference points in the direction the box has to move. GIoU lies in $(-1, 1]$ and equals IoU whenever $C = A \cup B$. Past 100 pixels IoU is a horizontal line and GIoU is still descending.

2.3 From one box to a whole detector

IoU scores one pair. A detector emits many boxes per image, most of them wrong, all carrying a score. Fixing a score threshold gives one precision and one recall — and throws away everything the ranking knew.

A detector is a ranking. Score it as one. Fix a class and an IoU threshold, sort every detection of that class over the whole corpus by score, and walk down the list: for each detection find the best *unmatched* annotation in the same image, and if that IoU clears the threshold it is a true positive and the annotation is now matched. That word *unmatched* is what makes a second box on the same bottle a false positive rather than a second success.

Precision and recall are unchanged from Lecture 3, computed cumulatively after each detection in the ranking. The recall denominator is fixed, so recall is non-decreasing; precision has both a numerator and a denominator moving, and on the person class it saws up and down exactly as Lecture 3's did.

Lecture 3 proved precision is not monotone, and detection repairs it: average precision is defined using the maximum precision at or above each recall level,

$$p_{\text{env}}(r) = \max_{\tilde{r} \geq r} p(\tilde{r}),$$

non-increasing by construction. Why a maximum rather than a smoothing? It has an operational reading — if you will accept recall r, here is the best precision available at that recall or better; it

is monotone by construction, so the area under it does not depend on where you sampled; and it is one line of code with no parameters.

$$\mathrm{AP} = \int_0^1 p_{\mathrm{env}}(r)\,\mathrm{d}r = \sum_k (r_k - r_{k-1})\, p_{\mathrm{env}}(r_k).$$

The sum only has terms where recall actually changed, which is exactly at the true positives — and *no threshold appears anywhere*. That is the whole point: the number does not depend on a cut-off nobody chose.

This *interpolated* AP, integrated at every recall level, is the detection convention, and it is what our notebook computes; COCO's own evaluator samples the same envelope at 101 recall levels, and agrees with ours to within 0.0014 below. Scikit-learn's `average_precision_score`, which Lecture 3 and the rest of the course use, does not interpolate: it sums $(r_k - r_{k-1})\, p(r_k)$ with the raw precision, which is the mean of the precision at each true positive. The two agree when the precision at each true positive is no higher than at the one before it; when it is higher, the interpolated AP is larger.

2.4 A mean of a mean of an area

The first mean is over classes: $\mathrm{mAP}(t)$ averages $\mathrm{AP}_c(t)$ over the classes present. Unweighted — a class with one annotation counts as much as person with 350. We average over the 73 classes appearing in our 128 images; the other 7 have no annotations here. *Which classes are in the mean is a decision, it changes the number, and papers do not always say which one they made.*

The second mean is over IoU thresholds, averaging the ten values from 0.50 to 0.95. At 0.50 a box that half-covers the object counts as correct; at 0.95 it must be very nearly exact. That average is COCO's headline number and what "mAP" means unqualified in a modern paper — and on our corpus it is 0.439, against 0.659 at the loosest threshold alone, with one number standing for all ten.

3 Running a detector, and reading it

The specification first: 128 JPEG files and one annotation file in; per image, boxes in corner form, category ids, scores, and one integer count out; nothing trained; and the *check* — every predicted box lies inside the image, scores are non-increasing, and the count equals the number of boxes kept. That last row is the one people skip, and it is the one that catches a corner-versus-size mix-up in a line.

Convert once, at the edge, and assert it

COCO gives x, y, w, h; everything else here is corners. The conversion happens once, on load, and then:

```
assert (b[:, 2] >= b[:, 0]).all(), "x2 < x1 — you read w as x2"
assert (b[:, 3] >= b[:, 1]).all(), "y2 < y1 — same bug"
```

Reading w as x_2 gives $x_2 < x_1$ for every box wider than its own left edge, which is almost all of them. Two lines catch it; nothing else will.

Failure condition — there is already a cut-off, and it is not yours

`box_score_thresh` defaults to 0.05, so anything below is never returned at all. `box_nms_thresh` defaults to 0.5, so near-duplicates are removed before you see them. `box_detections_per_img` defaults to 100 — and our busiest image returns exactly 100 boxes, so that cap is not hypothetical.
Three defaults you did not ask for.

The count is a function of the threshold: move the cut-off and the answer to the stakeholder's question changes by a factor of ten. We adopt 0.5, because it is the conventional reporting threshold *and because it was chosen before looking at the error curve*. The minimum of that curve sits at 0.75 with an MAE of 1.92, and adopting it would be choosing a hyperparameter on the data we then report — Lecture 2's error in a new costume. So it is quoted as a diagnostic and not as a result.

At score at least 0.5	Value
Count MAE	3.00
Signed count error	+2.41
Images counted exactly right	27 of 128
Images with too many boxes	83
Images with too few	18
Boxes kept in total	1,206

Two numbers, not one: the absolute error says how far off, the signed error says which way — and the cloud sits above the diagonal, so the failure mode is too many boxes. 3.00 against a baseline of 6.02, so the model halves the error of a system that never looks at the image. The same model with the threshold line deleted — every box it returns — scores 27.51: four times worse than predicting nothing at all.

3.1 Check your own implementation against someone else's

We derived AP here and implemented it ourselves, which is exactly the situation in which a result should not be believed.

mAP on our 128 images	Ours	Reference evaluator
at IoU 0.50	0.659	0.660
at IoU 0.75	0.475	0.476
averaged 0.50 to 0.95	0.439	0.440

Largest disagreement 0.0014, against the official COCO evaluator run on the same predictions by a library we did not write.

And now compare 128 images against 5,000

Our averaged mAP is 43.9 on 128 images. torchvision reports 37.0 for *the same Faster R-CNN weights* on all 5,000 images of `val2017`.
6.9 points higher, and nothing about the detector changed. The difference is the sample: 128 images, many of them easy, and 73 classes several of which have a single instance that happened to be found. This is what a small evaluation set does — and it does it in the flattering direction more often than not.

3.2 Suppression

The detector you ran had already suppressed near-duplicate boxes before you saw them, using IoU, at the `box_nms_thresh` of 0.5 you did not set.

Non-maximum suppression is greedy and per class: sort the candidates by score, keep the highest, delete every remaining candidate whose IoU with it exceeds the threshold, and repeat on what is left. It deletes information that cannot be recovered afterwards.

Failure condition — suppression merges real neighbours

Two genuinely different objects that overlap by more than the threshold become one box. Nothing raises, and the count comes out one short.

Use `batched_nms` rather than `nms`: a bottle overlapping a glass must not suppress it, because they are different classes and both are true. The suppression threshold is another knob: sweeping it from 0.1 to 0.9 takes the boxes kept per image from 14.6 to 134.1, and mAP at IoU 0.5 peaks around a threshold of 0.5. The default happens to be near the best value on our corpus — a fact about this corpus, not a law.

4 Tracking, and segmentation

4.1 Detection on video is not tracking

Run the detector on every frame and you have boxes, not the same bottle in frame 2 as in frame 1. Tracking asks for an identity that persists across frames, which turns each frame into an assignment problem: match this frame's detections to the existing tracks. The cheapest similarity between a track's predicted box and a new detection is IoU, so the derivation pays for itself a third time. A tracker, in four lines: predict where each existing track's box will be in this frame; compute IoU between every prediction and every new detection; solve the assignment, pairing them to maximise total IoU; and let unmatched detections start new tracks while unmatched tracks age and die.

Failure condition — identity carried by overlap alone

An object occluded for ten frames reappears somewhere its predicted box does not overlap. IoU is zero, the assignment finds nothing, and the object is issued a new identity — the same flatness, in a new place. The tracker does not report a lost track; it reports a new one, and the count it hands the stakeholder is wrong in a way the video makes obvious and the number does not.

4.2 Per-pixel prediction

A box was always an approximation. A bicycle's box is mostly not bicycle; a person sitting cross-legged fills perhaps half of theirs; and for a picking arm "somewhere in this rectangle" may not be good enough.

Semantic segmentation gives one class label per pixel, so two people standing together are one person region and you cannot count them. The architectural problem is that pooling destroyed the resolution the answer needs — Lecture 12's invariance arriving as a cost — and the repair is upsampling with skip connections: transposed convolutions that recover spatial detail from the earlier, higher-resolution layers.

Instance segmentation gives one binary mask per detected object. Mask R-CNN is Faster R-CNN with one extra head predicting a small mask inside each proposed box, and it is two lines away from what you already ran; its `masks` output has shape $(N, 1, H, W)$, soft masks in $[0, 1]$. Semantic and instance segmentation differ in exactly one place — whether two touching objects of the same class are one region or two — and only the instance output can answer "how many".

Scoring a mask is the same argument again: mask AP replaces box IoU with mask IoU, intersecting pixel sets instead of rectangles, and everything above — the ranking, the envelope, the area, the two means — is unchanged. For the Mask R-CNN weights, torchvision reports 34.6 mask mAP against 37.9 box mAP on the full 5,000 — which is what "a mask is harder than a box" means numerically. (That 37.9 is Mask R-CNN's own box score; the Faster R-CNN we ran is the 37.0 above.) We do not measure mask AP on our corpus; the pixel-level annotations are in the file and the code is an exercise.

5 The notebook

What to change

Move one predicted box further and further from its match, and watch IoU sit at exactly zero the whole way. That flat zero is the vanishing gradient the derivation predicts — and seeing it as a constant rather than as a small number is the difference between "the gradient is very small", which is wrong, and "the gradient is zero", which is the point.

6 Design exercise: a parking-analytics company's occupancy product

The lecture closes with a design exercise in the manner of an ML system-design interview, on the ML system behind a company's product, whose answer is a set of boxes and a count.

The task. A parking-analytics company flies drones over car parks and sells the result to operators and cities: for every image, how many vehicles are parked and where each one is, as a count on a dashboard and a set of boxes through an API, so that an operator can check bays against bookings and a city can plan. A list of the categories present is no use; the feature at the centre is the detector behind the count. The business measures the contracts it keeps, the accuracy clause written into each one, and the cost of a count a customer proves wrong. Given: a few hundred drone images across a dozen customer sites, each vehicle boxed and classed — car, van, lorry, motorcycle — by a labelling contractor, and a pretrained detector. A few hundred vehicles on a full site, images several thousand pixels across, a flight over every site several times a day. The output feeds the operators' occupancy dashboard, their booking reconciliation, and the cities' planners.

1. **Frame it.** Output per image: boxes in corner form, a class and a score for each, and one integer count. A predicted box is correct when its IoU with a still-unmatched annotation of the same class clears a threshold stated in advance; the count is a separate number with its own error, absolute and signed. Fix the box convention once, at the edge: the contractor's tool exports a corner and a size, the detector returns two corners, so convert on load and assert that the second corner lies below and to the right. *Pitfall:* the count alone as the metric. A detector that misses the cars under the trees and doubles the ones in the sun returns the right total: the count error is near zero, the boxes are wrong, and the operator's bay check fails on the first contract. Score the boxes by IoU matching, and the count beside them with its sign.

2. **The annotations, and what the contractor's boxes are worth.** A box drawn on a drone image is a judgement: does it include the shadow, the open door, the roof rack; is a car half under a canopy annotated at all; is a trailer a vehicle. Write the rule into the contractor's brief, give the same images to two annotators, and measure their IoU on the same vehicles. *Pitfall:* a threshold the annotators cannot meet. Two careful people boxing the same car from the air disagree by more than the strict thresholds allow, so at those thresholds a perfect detector is scored wrong by the annotation's own noise, and the headline mean over thresholds is capped by the annotators rather than by the model. Report the inter-annotator IoU beside the evaluation, take the loose threshold as the operating number, and send the disagreements back to be resolved before anything is scored.

3. **Split by site, and keep the empty tarmac in.** Hold whole customer sites out, as every new customer's site will be: tiles from the same flight share the vehicles, the light and the paint, so a split by tile scores memory. Most of every image is background — tarmac, painted bays, drain covers, shadows — and the evaluation set must hold it in its true proportion; report the baselines of no boxes at all and one box per tile. *Pitfall:* the background as the dominant class, dropped. A first attempt keeps the tiles with vehicles in them and discards the empty ones as "nothing to annotate"; the false positives on empty bays — the painted markings and drain covers a detector likes — are never counted, and the dashboard overstates occupancy on every quiet site. Keep the empty tiles, annotated as empty, and report the boxes returned

on them.

4. **Representation: the vehicle at the detector's input size.** The detector's transform resizes the image to its own working size: a drone frame several thousand pixels across is shrunk several-fold, and a car that was forty pixels long becomes a handful. The backbone's pooling discarded resolution on purpose; a detector keeps what it can, but not an object a few pixels across. *Pitfall:* small objects at that resolution. Resize the whole frame and the vehicles at the far edge of an oblique shot vanish below what the feature maps can resolve; the AP is fine on the near, large vehicles and zero on the far, small ones, and the mean hides the second. Tile the frame at native resolution, or fly lower, and check it with an AP per object size.

5. **The model: the detector as it ships, and its three defaults.** Load the pretrained detector in `eval()` mode; nothing is trained; run the corpus under inference mode. Three settings nobody chose: a score cut-off, a suppression threshold, and a cap on detections per image. A full site holds more vehicles than the default cap, so the count saturates: raise it before anything is measured. Pick the score cut-off before looking at the error curve, and write it into the contract; the minimum of that curve on your own images is a diagnostic, not a result. *Pitfall:* non-maximum suppression. Greedy, per class: a car parked bumper to bumper with the next, seen obliquely, overlaps it by more than the threshold and the second is deleted; the count comes out one short per row, and nothing raises. Suppress per class, raise the threshold, and check it on the packed rows — the sweep on your own corpus says what it does here, not in general.

6. **Failure condition: where it fails, by size and by place.** The detector misses what is small, shaded or partly covered: the near half of an oblique frame is easy and the far half is not, and one AP over the whole site averages them. Bucket the annotations by box area and compute AP for each bucket, as the reference evaluator does, and the count error by region of the frame. *Pitfall:* evaluation per object size, skipped. The single AP is carried by the large near vehicles, the small-object AP is a fraction of it, and the customer's complaint — the far rows are always under-counted — is invisible in the number reported to them. Report AP per size bucket and the signed count error per region, and fix the small ones at the input, not the threshold.

7. **Evaluation: what the one number hides.** mAP is a mean over classes of a mean over thresholds: the motorcycle class, with a handful of annotations, counts as much as cars with thousands, and the strict thresholds are the ones the annotators cannot meet. Check your own implementation against the reference evaluator on the same predictions before quoting it, and say the sample size beside every number, because a few hundred images flatter a detector, in the easy direction more often than not. *Pitfall:* a headline from your own evaluator on a small set. A mAP computed by code you wrote, on a few hundred images, several points above the published figure for the same weights, goes into the sales deck as a gain and the accuracy clause is signed against it; the gain is the sample, and possibly the evaluator. Match the reference to within rounding first, then state the sample size, the operating threshold and the per-class APs with every figure.

8. **In production: tiles, seams, and the barrier count.** A site image is split into overlapping tiles at native resolution, each run through the detector, and the boxes mapped back to frame coordinates. The operator's barrier count is a free ground truth for the total: a new detector runs in shadow beside the old one on every flight, its signed error against the barrier is read

before it takes over, and the analyst on call watches that error for a step — new paint, snow, a canopy built. *Pitfall:* tiled inference and its seams. A car cut by a tile edge is two half-cars, one in each tile, and both are detected — or neither, if each half is too small to clear the cut-off — so the count double-counts along every seam. Overlap the tiles by more than a vehicle's length, map the boxes to frame coordinates, run suppression once more across the whole frame, and then check the count against the barrier's.

9. **Ship it.** Boxes to corners once, asserted → annotation rule, inter-annotator IoU → split by site, empty tiles kept → native-resolution tiles → detector as shipped, cap raised, cut-off fixed in advance → suppression per class → AP per size, signed count → evaluator checked against the reference → tiles merged, suppressed across seams → shadow run against the barrier count.

7 Exercises

The five set at the end of this lecture's deck, in the style of the written paper. The solutions are on Lecture 15's deck and in the exercise book.

1. Two boxes are disjoint and moved further apart. State what IoU and its derivative do, and why no learning rate repairs it. [5]

2. A detector counts objects with MAE 3.00 at the conventional score cut-off of 0.5. A sweep over the same 128 images finds MAE 1.92 at 0.75. State why 1.92 must not be reported as the result, and how it may be quoted. [5]

3. Average precision is defined with a maximum inside it. State what that maximum repairs, and which earlier lecture proved the problem. [4]

4. mAP is a mean over classes of a mean over IoU thresholds. Give one thing each averaging hides. [4]

5. Your mAP over IoU 0.50 to 0.95 on 128 COCO images is 43.9; torchvision reports 37.0 for the same weights on all 5,000. Name the check that rules out your own evaluator first, and then the cause of the gap. [4]

Summary

A label is not an answer when the question is *where* and *how many*, and a network trained to be invariant to position cannot supply either. IoU is the essentially unique quantity meeting the four requirements — and it is exactly zero for every disjoint pair, so as a loss it offers no gradient however far apart the boxes are. Not a small one: none, because the clamp makes it constant. GIoU repairs that by adding a term that keeps moving.

Average precision is built by matching a ranking against annotations, and its definition contains a maximum for one reason: precision is not monotone, which Lecture 3 proved, and the envelope is the repair. mAP then averages over classes and over ten IoU thresholds — a mean of a mean of an area, where our own 0.439 averages thresholds whose loosest alone gives 0.659.

Our implementation agrees with the official evaluator to 0.0014, which is why it can be quoted. And it reports 43.9 where torchvision reports 37.0 for the same Faster R-CNN weights: 6.9 points, entirely from evaluating on 128 images instead of 5,000, in the flattering direction. The closing design exercise built the occupancy product a parking-analytics company sells to operators and cities in nine steps — boxes scored by IoU beside a signed count, an annotation rule measured between annotators, empty tarmac kept in the evaluation, tiles at native resolution for the small vehicles, suppression checked on packed rows, AP per object size, the evaluator matched to the reference with the sample size stated, and tiles merged with suppression across the seams — each step named for the pitfall it avoids.